CITY COLLEGE

Approved by AICTE, New Delhi. Recognized by Govt. of Karnataka. Affiliated to Bangalore University, accredited by NAAC with "B+" Grade No.27/2, 33rd Cross, 2nd Main Rd, 7th Block,Jayanagar, Bengaluru, Karnataka 560070 Phone: 6364914155 / 56,

+918026649277

Website: www.citycollege.ac.in Email: admin@citycollege.ac.in

Project Report On

Microsoft AI Security Key Exposure Incidents (2024) Cloud and AI security topic. Modern and relevant.

Submitted in partial fulfilment of the requirement for the project work for the degree of

(BACHELOUR OF COMPUTER APPLICATION) BCA

2025–26

BANGALORE UNIVERSITY

Under the Guidance Of

Ms. Chandana Khatavakar

Assistant Professor

Department of Computer Application

Submitted by

YASHASWINI D P (U03BX24S0114)

CITY COLLEGE, JAYANAGAR, BENGALURU

DEPARTMENT OF COMPUTER APPLICATION

CERTIFICATE

This is to certify that the project synopsis entitled “Microsoft AI Security Key Exposure Incidents (2024) Cloud and AI security topic Modern and relevant” is submitted by Ms Yashaswini D P(USN: U03BX24S0114) in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from Bangalore University for the academic year 2025–2026.

The project synopsis has been prepared under the guidance and supervision of the undersigned. The synopsis is submitted for academic evaluation in accordance with the applicable department and university guidelines.

_____	_____	_____
Project Guide	**Head of the Department**	**Principal**
Ms. Chandana Khatavakar	Dr. Nageshwari	City College
Assistant Professor	Department of Computer Application	Jayanagar, Bengaluru

Place: Bengaluru

Date: 11-10-2026

Abstract

Artificial intelligence (AI) and cloud computing have become important technologies for modern organizations. They support data storage, application development, research, collaboration, and intelligent services. However, these technologies also introduce security risks when sensitive information, access credentials, or cloud permissions are not properly protected.

This case study examines Microsoft-related cloud and AI security risks through two documented security events from 2024. The first is the exposure of data associated with Microsoft's AI research team through an incorrectly configured Azure Storage Shared Access Signature (SAS) token, publicly reported by cloud security researchers in 2023. The second is the Midnight Blizzard attack against Microsoft's corporate systems, detected in January 2024 and publicly disclosed by Microsoft. These are separate events and should not be described as one incident.

The AI research data exposure demonstrated how an overly permissive or improperly handled access token could make cloud-hosted files accessible to unintended parties. The Midnight Blizzard incident demonstrated how attackers could exploit a compromised legacy test account to access certain corporate email accounts. Together, these events highlight the importance of identity security, access control, secret management, continuous monitoring, and timely incident response.

This study discusses the incident background, attack methods, potential root causes, impact, detection, response, lessons learned, and recommended security measures. It also explains how organizations can reduce similar risks by applying least-privilege permissions, multifactor authentication, secure credential storage, regular security audits, and automated detection tools.

The case study is based on publicly documented information and credible cybersecurity sources. It does not claim that the two events were connected or that the exposed AI research data was necessarily used in the Midnight Blizzard attack. The central conclusion is that AI innovation must be supported by strong cloud security practices, clear accountability, and continuous risk assessment.

Introduction

Artificial intelligence and cloud computing are transforming the way organizations develop software, analyze information, and deliver digital services. Microsoft uses cloud technologies such as Azure to support applications, data storage, AI development, and enterprise operations.

Despite these advantages, cloud environments can be exposed to security problems when access tokens are misconfigured, credentials are compromised, or accounts have excessive permissions.

An access key or token is a digital credential that allows a user or application to access a protected resource. If a credential becomes publicly accessible and remains valid, an unauthorized person may be able to use it.

This case study focuses on Microsoft's AI-related data exposure and its separate January 2024 corporate security incident. Both illustrate the importance of protecting digital identities and sensitive information.

Objectives:

- Understand cloud and AI security risks.
- Examine documented Microsoft-related incidents.
- Identify possible weaknesses in security controls.
- Study incident detection and response.
- Recommend measures to improve organizational security.

Incident Description

The main AI-related data exposure was reported by Wiz Research on September 18, 2023. Microsoft AI researchers had shared a link to training data in a public GitHub repository. The Azure Storage Shared Access Signature (SAS) token associated with the link was configured with broader access than intended.

According to Wiz, this configuration exposed approximately 38 terabytes of additional private data, including workstation backups, secrets, private keys, passwords, and more than 30,000 internal Microsoft Teams messages.

The issue involved an access-control misconfiguration rather than a confirmed attack exploiting an AI model vulnerability. The report demonstrated how a cloud-sharing feature

intended for legitimate research can create a significant data exposure risk when permissions are too broad.

Separately, Microsoft disclosed on January 19, 2024, that the Midnight Blizzard threat actor had accessed a limited number of corporate email accounts after compromising a legacy test account. Microsoft detected the activity on January 12, 2024.

These events provide two different lessons: carefully control cloud data-sharing permissions, and strengthen identity security across corporate systems.

Attack Type / Vulnerability

The AI research data exposure was caused by a cloud-storage access configuration problem. Azure SAS tokens allow users to share selected storage resources without making an entire storage account public.

In this case, the token provided access to a broader scope of data than the researchers intended to share.

Key security weaknesses:

1. **Excessive permissions**: The token allowed access beyond the intended research files.
2. **Credential exposure:** Sensitive information was present in the exposed workstation backups.
3. **Cloud configuration risk:** A valid sharing mechanism was configured unsafely.
4. **Sensitive-data concentration:** Large volumes of data were stored together.
5. **Potential secret misuse:** Exposed credentials can create further risks if they remain valid.

The separate Midnight Blizzard incident involved password spraying against a legacy, non-production test account. It was an identity compromise, not the same vulnerability as the SAS-token misconfiguration.

Attack classification: Accidental cloud data exposure and, separately, a targeted corporate account compromise.

Root Cause Analysis

The main root cause of the AI research exposure was an incorrectly scoped Azure SAS token. The token was designed to share research files, but its permissions covered additional private data.

The incident highlights several underlying control weaknesses:

- **Incorrect permission scope**: Access was not restricted to only the required files.
- **Insufficient validation**: The sharing configuration did not prevent the unintended exposure before the link was published.
- **Secrets stored with other data:** Backups contained credentials and other sensitive information.
- **Long-lived access risk:** SAS tokens can be configured with different expiration periods; overly long validity increases exposure risk.
- **Limited separation:** Sensitive backups and research data were accessible through the same broad sharing mechanism.

For the separate Midnight Blizzard incident, Microsoft reported that attackers used a password-spraying technique against a legacy test account and leveraged its permissions.

Root-cause conclusion: Secure cloud systems require correct configuration, least-privilege access, protected credentials, and consistent security standards for both production and legacy environments.

Impact Analysis

The AI research data exposure created significant confidentiality risks because a large volume of private information became accessible through a sharing link.

1. Data confidentiality: The exposed material reportedly included private messages, workstation backups, passwords, and private keys.

2. Credential security: Secrets found in backups could potentially enable access to other systems if they remained valid.

3. Research security: Exposure of AI research files and related development information can create risks to intellectual property and research integrity.

4. Privacy concerns: Internal communications and employee-related information may contain sensitive business or personal details.

5. Organizational reputation: Security incidents can reduce confidence in an organization's ability to protect sensitive information.

In the separate Midnight Blizzard case, corporate email correspondence was accessed and some emails and attachments were taken. Microsoft reported no evidence at that time that the attackers accessed customer environments, production systems, source code, or AI systems.

The public reports establish the nature of the exposures, but they do not prove that every exposed secret was misused or that all possible consequences occurred.

Overall impact: The incidents demonstrate how weaknesses in cloud permissions and account security can create substantial organizational risk.

Detection and Response

In the AI research exposure case, Wiz Research identified the misconfigured storage link while examining publicly accessible cloud storage configurations. The researchers reported the problem to Microsoft, and Microsoft addressed the exposure.

The discovery demonstrates the value of independent security research, responsible disclosure, and continuous cloud configuration assessment.

In the separate Midnight Blizzard incident, Microsoft's security team detected suspicious activity on January 12, 2024. Microsoft investigated the intrusion, disrupted malicious activity, and worked to prevent further access. The company publicly described its findings on January 19, 2024.

Important response activities include:

1. Identify the affected resource or account.
2. Restrict or remove unauthorized access.
3. Revoke or rotate exposed credentials.
4. Review access logs to determine what was accessed.
5. Investigate whether related accounts or resources were affected.
6. Notify relevant stakeholders when required.
7. Apply corrective controls and monitor for recurrence.

Key lesson: Rapid containment must be followed by investigation and long-term remediation. Removing a public link alone may not be sufficient if a credential remains valid.

Security Measures and Mitigation

Organizations can reduce cloud and AI security risks by adopting the following measures:

1. Least-privilege access: Grant users and applications access only to the resources necessary for their work.

2. Secure token configuration: Limit SAS tokens to specific files, minimum permissions, and short validity periods.

3. Credential rotation: Immediately revoke or rotate exposed keys, passwords, and tokens.

4. Secure secret storage: Use managed secret stores such as Azure Key Vault instead of placing credentials in code, public repositories, or backups.

5. Multifactor authentication: Protect user accounts with MFA and stronger phishing-resistant methods where available.

6. Continuous monitoring: Monitor cloud storage, identity activity, access changes, and unusual downloads.

7. Data classification: Identify confidential data and apply stronger controls to sensitive information.

8. Security testing: Regularly review storage permissions, access policies, and cloud configurations.

9. Secure development practices: Review repositories and research workflows before publishing code or data.

These controls reduce the probability of accidental exposure and help limit damage when credentials are compromised.

Case Study Diagram / Infographic

Use this diagram on your slide as a flowchart. It summarizes the AI data exposure, not the separate Midnight Blizzard attack.

AI research files prepared for sharing

Researchers publish a download link in a GitHub repository

Overly broad SAS token

The token grants access beyond the intended research files

Private data exposed

Wiz reports approximately 38 TB of additional data was accessible

Detection and remediation

Report the exposure, restrict access, rotate secrets, and audit permissions

Source: Wiz Research, “38TB of data accidentally exposed by Microsoft AI researchers” (2023).

Lessons Learned

The incidents provide several important lessons for organizations adopting cloud computing and AI technologies.

Lesson 1: Cloud-sharing settings matter. A storage account can remain private while an overly permissive token grants unintended access.

Lesson 2: Credentials must be treated as sensitive data. Passwords, keys, and tokens should not be stored in research backups or exposed repositories without appropriate protection.

Lesson 3: Least privilege is essential. Permissions should match the specific task and should not provide unnecessary access to entire storage accounts.

Lesson 4: Legacy accounts require protection. Old test environments and unused accounts can become entry points for attackers.

Lesson 5: Monitoring supports early detection. Cloud configuration scanning, identity monitoring, and audit logs can reveal risky access before damage expands.

Lesson 6: Incident response must be comprehensive. Organizations should contain exposure, rotate credentials, investigate access, and verify that remediation is effective.

Lesson 7: AI security is also data security. AI development relies on datasets, code repositories, cloud storage, and credentials. Each component needs appropriate protection.

The central lesson is that secure technology requires secure configurations, disciplined access management, and ongoing review.

Recommendations

Based on the findings, the following recommendations are proposed for organizations using cloud services and AI development environments.

1. **Apply least privilege**: Restrict access to individual files or containers rather than entire storage accounts whenever possible.
2. **Set token expiry**: Use short-lived SAS tokens and regularly review active sharing links.
3. **Use managed identities**: Reduce reliance on long-lived credentials when secure identity-based authentication is available.
4. **Scan for exposed secrets**: Integrate automated secret detection into GitHub repositories and development pipelines.
5. **Protect legacy accounts**: Remove unused accounts, enforce MFA, and review permissions for test environments.
6. **Enable cloud monitoring**: Collect and analyze logs for unusual access, permission changes, and large downloads.
7. **Encrypt sensitive data:** Apply appropriate encryption and secure key management.
8. **Conduct regular audits:** Review cloud configurations, token permissions, and data-sharing practices.
9. **Create an incident-response plan**: Assign responsibilities for containment, investigation, communication, and recovery.
10. **Train employees and researchers**: Teach safe data sharing, credential handling, and responsible publication practices.

Organizations should prioritize high-risk findings, document corrective actions, and verify that each security improvement is working as intended.

Conclusion

The Microsoft AI research data exposure and the separate Midnight Blizzard corporate intrusion demonstrate the importance of strong cloud and identity security.

The AI-related exposure showed how an incorrectly configured SAS token could make private information accessible beyond the intended scope. The January 2024 Midnight Blizzard incident showed how a compromised legacy account could provide an entry point to corporate email systems.

Although the incidents were different, both emphasize the need for secure authentication, least-privilege permissions, protected secrets, monitoring, and effective incident response.

Organizations developing AI systems should protect not only their models but also the datasets, repositories, storage accounts, identities, and credentials that support them. Security checks should be integrated into everyday development and publishing workflows rather than added only after an incident occurs.

In conclusion, AI innovation and cloud security must develop together. Regular audits, responsible data sharing, secure defaults, and continuous monitoring can help organizations reduce exposure and build greater trust in modern digital services.

References

Use these credible sources in your PPT. Keep the links clickable in PowerPoint.

1. Wiz Research (2023)

38TB of data accidentally exposed by Microsoft AI researchers.

Primary technical investigation of the AI research data exposure.

2. Microsoft Security Response Center (2024)

Microsoft Actions Following Attack by Nation State Actor Midnight Blizzard.

3. Cybersecurity and Infrastructure Security Agency (2024)

Emergency Directive 24-02: Mitigating the Significant Risk from Nation-State Compromise of Microsoft Corporate Email System.

4. Microsoft Security Blog (2024)

Securing our future: September 2024 progress update on Microsoft's Secure Future Initiative.

5. Microsoft Learn

Secure Future Initiative overview.

6. Microsoft Learn

Protect identities and secrets in the Microsoft Secure Future Initiative.

The first three references support the incident descriptions; the remaining references support the security improvements and recommendations.

Disclaimer

This case study has been prepared for academic and educational purposes using publicly available information from Microsoft, Wiz Research, CISA, and Microsoft Learn.

The AI research data exposure reported by Wiz in September 2023 and the Midnight Blizzard incident disclosed by Microsoft in January 2024 are separate events. They are discussed together only to examine broader cloud and identity security lessons. No claim is made that the incidents were connected.

The analysis of contributing factors and recommendations is educational and should not be interpreted as an official Microsoft investigation report. All factual claims should be checked against the cited sources before publication.